2.4 Dark Field Microscope

This is used for examining live unstained microorganisms. The distinct feature is the dark field condenser that contains an opaque disc. The disc blocks direct entry of light to the objective lens. The light rays reflected off the specimen enter the objective lens and in the absence of direct background light, the specimen appears light against a dark background (Figure 2.9). The microbes are visualized as halos of bright light against the darkness, as stars are observed against the night sky (Figure 2.10).

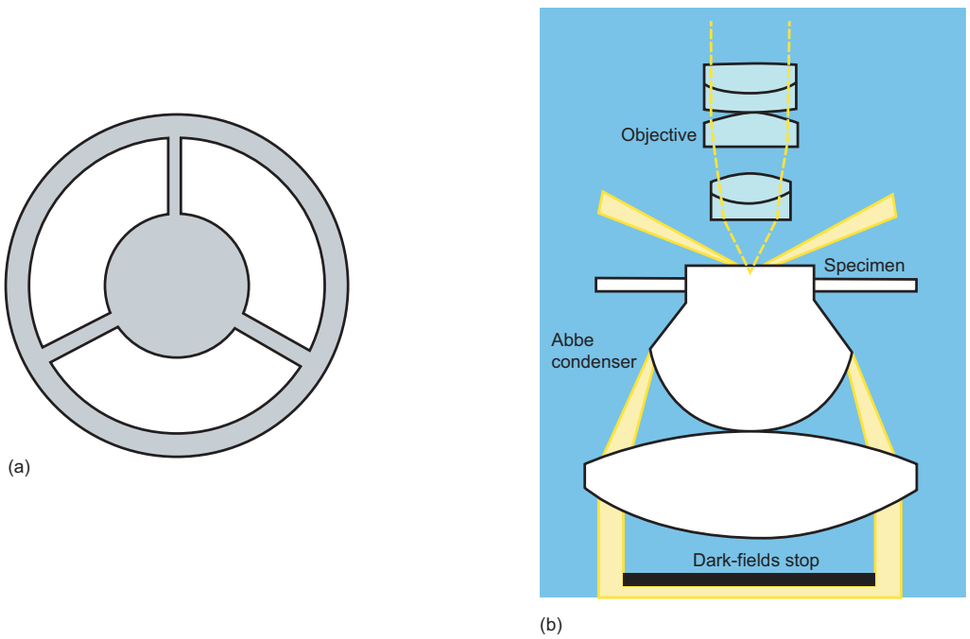


Figure 2.9: Dark Field Microscopy. The simplest way to convert a microscope to dark field microscope is to place. (a) a dark field stop underneath (b) the condenser lens system

Infobits

Compound microscope (also known as light microscope) produces a mono (2D) image and stereo microscope produces stereo (3D) image. 'Upright' life science microscopes are the most numerous of all microscopes. An inverted microscope is the kind of microscope that views objects from an inverted position. Digital microscopes are becoming widespread. These provide simple image and are convenient for electronic image capturing.

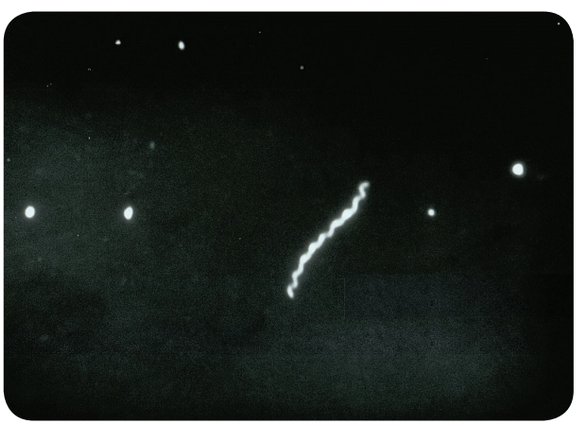


Figure 2.10: Dark field observation of bacteria *Treponema pallidum* specimen from a patient with Syphilis

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